

Grid Completion

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to create an $n \times n$ grid whose each row and column has exactly one A and B. Some of the characters have already been placed. In how many ways can you complete the grid?

Input

The first input line has an integer n : the size of the grid.

After this, there are n lines that describe the grid. Each line has n characters: . means an empty square, and A and B show the characters already placed.

You can assume that every row and column has at most one A and B.

Output

Print one integer: the number of ways modulo $10^9 + 7$.

Constraints

- $2 \leq n \leq 500$

Example

Input	Output
5AB. B..... ...A.	16